

The Structure of Belief Networks in Mass Publics

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Abstract

In “The Nature of Belief Systems in Mass Publics,” Philip Converse famously provided an account describing the mass public as non-ideological, exhibiting little to no constraint among issue opinions. Still, since the publication of his famous 1964 essay, a great deal has changed. The dominant political cleavages have shifted from economic to social issues. Party cues have largely become much clearer. Partisan and ideological sorting has reached an all-time high. Sources of national political information have proliferated. Has any of this changed the relative constraint of mass belief systems? To shed light on this question, we use ANES data spanning a 32-year time period to conduct a series of network analyses aimed at explicating the structure of the mass public’s belief systems. Using measures of network clustering—network density and average path length—we show that constraint remained relatively stable in the 1990s and 2000s, with a substantial increase in issue constraint occurring in 2012 that persisted through 2024. We subset these analyses by political knowledge and find that this change is largely driven by low-information rather than high-information respondents. By 2024, the differences in constraint across these groups attenuate significantly. Furthermore, using measures of network centrality, we find that racial resentment, authoritarianism, and partisanship are strikingly strong organizers of belief systems. These attitudes’ influence over mass belief systems has markedly increased from 1992 to 2024. We again subset these analyses by respondents’ knowledge and find that high-knowledge respondents are much better at using political concepts such as ideology in structuring political opinions compared to low-knowledge respondents. Additionally, this difference is present for partisanship in 1992 but not in 2024, while racial resentment and authoritarianism maintain relative hegemony by 2024 regardless of information levels.

Introduction

Converse’s 1964 essay is perhaps the most well-known among political scientists. In it, he sought to characterize the nature and structure of mass belief systems, which he defined as “a configura-

tion of ideas and attitudes in which the elements are bound together by some form of constraint or functional interdependence” (p. 3). According to Converse’s analyses, the masses’ belief systems were generally unconstrained, unorganized, and certainly were not structured by ideological thinking. Though elites’ belief systems were certainly more organized than the masses’, this signal went largely unnoticed by the general public such that very little constraint, defined as “the success we would have in predicting, given initial knowledge that an individual holds a specified attitude, that he [or she] holds certain further ideas and attitudes” (Converse, 1964, p. 3), was exhibited for most issues. Thus, the “diffusion” of the elites’ relatively constrained belief systems to the masses was largely non-existent.

However, much has changed since Converse’s essay was published. The Civil Rights Movement upended the dominant political cleavages of the first half of the 20th century, albeit gradually (Carmines and Stimson, 1989). Today, elites are more polarized than at any time since the Civil War (Hare and Poole, 2014). As partisan elites polarized, so too did the masses (Hetherington, 2001), leading to sharper partisan divisions (e.g., Kalmoe and Mason, 2022). Moreover, the information environment for receiving elite cues has shifted dramatically. Rapid technological changes, such as the Internet, have led to a greater abundance of national political information (Hopkins, 2018). Such information is more easily accessible than ever before. Has any of this changed the levels of issue opinion constraint within mass belief systems?

Though there is some suggestive evidence that issue opinion constraint, sometimes referred to as ideological consistency, in mass belief systems may have increased recently (Simas, 2023; Kozlowski and Murphy, 2021), the nature and extent of this constraint is hotly contested (e.g., Kinder and Kalmoe, 2017). Studies that seek to characterize constraint changes over broad swathes of time often suffer from calculating and representing correlations effectively without resorting to reporting average correlation coefficients (Baldassarri and Gelman, 2008; Kinder and Kalmoe, 2017). Moreover, there is much more disagreement over what exactly is structuring belief systems: partisanship, ideology, group-based attitudes, values, or socialized attitudes, to name a few. Quantitatively characterizing belief systems is inherently a difficult task. Famously, Converse began his 1964 essay with the following qualification: “Belief systems have never surrendered easily to empirical study or quantification. Indeed, they have often served as primary exhibits for the doctrine that what is important to study cannot be measured and that what can be measured is not important to study” (p. 1).

Advances in network analysis allow us to overcome these difficulties and empirically represent the structure of belief systems as *belief networks*. By conceiving of belief systems in this way, we can formally represent the “functional interdependence” explicated by Converse and others to underpin what can be described as a belief system. We use a novel approach to do so: psychometric network analysis. This method allows us to more accurately represent that constellation of beliefs which structures mass belief systems (Converse, 1964). Network analyses have increasingly garnered attention in other social science fields such as psychology (e.g., Dalege et al., 2016; Brandt, 2020; Borsboom et al., 2021; Briganti et al., 2024) and sociology (Boutyline and Vaisey, 2017; Della-Posta, 2020), but this approach has not been used widely in the study of public opinion in political science, specifically (with the notable exception of Fishman and Davis, 2022). Moreover, to our knowledge, there has only been one belief network analysis that has examined temporal changes

in belief networks over time; however, this study is limited to studying changes in political beliefs through the 2008 election cycle alone (Fishman and Davis, 2022).

We regard the growing body of work representing political belief systems via psychometric network analyses as invaluable yet incomplete. We argue that network analysis techniques recently developed (Epskamp et al., 2018; Isvoranu and Epskamp, 2023) and popularized (e.g., Brandt et al., 2019) can provide new insights into age-old questions of how belief systems in the United States are structured and how they might have changed in their relative constraint over time.

To investigate these questions, we utilize 21 survey questions that were consistently asked in nine American National Election Studies (ANES) cross-sectional surveys from 1992 to 2024. In doing so, we are able to characterize the structure of mass belief systems over a time period spanning 32 years of political history using psychometric network analysis. Using network analysis metrics such as average path length and network density, we show that while constraint—the interdependence among various issue positions—remained relatively stable for much of the time period analyzed, issue opinions have become more interrelated in recent years: Mass belief systems from 2012 to 2024 exhibited significantly more constraint than belief systems in the 1990s and 2000s. Moreover, the constraint continues to increase, albeit moderately, in the 2012-2024 period. These results suggest that belief systems today appear to be much more tightly organized, more “functionally interdependent,” than they were in the past. We further find that the increase in constraint we observe is heterogeneous by information levels. Our results indicate that the relative clustering of issue opinions over the past several decades can largely be attributed to low-knowledge respondents rather than high-knowledge respondents. Though both have seen significant increases in constraint from 1992 to 2024, the constraint differences between the two groups appear to have greatly attenuated in the most recent years of our analyses.²

²Again, we urge caution on two points. We are still determining the best measures to investigate this question. Our original measure, average path length, and our more recent measure, network density, both suggest the same findings, but both have limitations. Another issue is the appropriate cutoff point for categorizing respondents as high- or low-knowledge. Our main analyses present results using the top and bottom 30th percentiles, respectively, as the cutoffs. Upon closer inspection, we found that, though we use the same questions over time, the raw number of correct and incorrect responses for those categories fluctuates within these percentile ranges, potentially influencing the results. Thus, Figures 16 and 17 in the appendix reproduce our results using the strictest cutoffs possible—with high-knowledge being respondents who answered all knowledge questions correctly and low-knowledge being respondents who answered all knowledge questions incorrectly. Using these cutoffs, the results for high- and low-knowledge differences attenuating over time do not emerge in as stark a contrast. We are still weighing what method is best and

If constraint has increased during this time period, when our dataset begins, the question remains: What specific attitudes are structuring mass belief systems, and has this changed over time? We leverage the available ANES data to preliminarily investigate which attitudes are more central, i.e., structuring, constraining forces, within belief networks using node centrality. We calculate results using strength centrality, the most stable and reliable of network node centrality measures (Bringmann et al., 2019; Isvoranu and Epskamp, 2023), and find that racial resentment and authoritarianism have increased significantly in their relative influence on other political opinions and attitudes in the network. Partisanship, egalitarianism, and moral traditionalism remain relatively constant over time, while symbolic ideology sees an uptick in strength centrality, rivaling that of partisanship. Following our constraint analyses, subsetting by political knowledge shows that use of political concepts is vastly different between low- and high-knowledge respondents in 1992 and 2024, though these differences attenuate in later years. Our results provide suggestive evidence that socialized attitudes regarding race and authority-mindedness are increasingly important in structuring political belief networks.

In sum, this paper provides valuable insight into the age-old question of what structures belief systems and how we might best model constraint. We use a unique approach to do so: a series of network analyses spanning 32 years. Our results showing convergence in issue constraint among high- and low-information respondents have significant implications for democracy. Furthermore, we present a preliminary investigation into which attitudes might be most influential in structuring mass belief systems over time, representing a novel analysis that looks at multiple attitudes that could plausibly be constraining mass belief systems. Finally, we hope this research endeavor encourages more scholars to adopt network analysis techniques in future studies seeking to characterize mass belief systems over large time periods.

Constraint in Mass Belief Systems

Converse's findings revealed the public's knowledge of political matters was quite low. Though elites tended to have moderately high levels of constraint among issue opinions, very few among the public maintained a sophisticated view of politics that could be described as an "ideology" or anything near a coherent "belief system" as he defined it. A belief system, according to Converse, is "a configuration of ideas and attitudes in which the elements are bound together by some form of constraint or functional interdependence" (p. 3). By constraint, he means "the success we would have in predicting, given initial knowledge that an individual holds a specified attitude, that he [or she] holds certain further ideas and attitudes" (p. 3).³

Of course, Converse used the term belief system rather than ideology because of the vast ambiguity that the term engendered, even among political scientists. It appears this ambiguity continues to this day. Depending on the context, scholars refer to ideology as symbolic identification along

whether we ought to present multiple cutoffs and allow readers to form their own conclusions.

³Converse described constraint as either "static" or "dynamic" (p. 3). When we refer to constraint, we are most closely adhering to Converse's definition of static constraint. Our data rely on cross-sectional observational analyses of ANES data over a 32-year timespan, and we do not causally test whether changes in one idea-element precipitate subsequent changes in other idea-elements. See Fishman and Davis (2022) for network analysis more closely approximating dynamic constraint using 10 waves of panel data over the course of the 2008 election cycle.

a liberal-conservative continuum, as symbolic identification outside of that continuum, as a set of values that create an ideology, as a set of worldviews that undergird ideology (which is amenable to changing political environments), or as a set of issue opinions that are intercorrelated, either within a certain policy domain or not. It is the latter that we focus on here to best approximate Converse's original conception of belief systems. Operational ideology, i.e., opinions or attitudes that are intercorrelated (or not) across policy domains, is our first concern, though we turn to symbolic ideology in the latter half of the paper.

Surprisingly little research has been conducted in a way that characterizes generalized issue constraint across policy domains over time, which is critical for understanding the trends in potential changes within mass belief systems. Look too narrowly, and one might misattribute mass scale changes after only a few election cycles (e.g., Nie and Andersen, 1974; Nie et al., 1979; Sullivan et al., 1978; Kinder and Kalmoe, 2017). Perhaps this is due to challenges in representing results efficiently; as Baldassarri and Gelman (2008) put it, "One can have a first, suggestive idea of the trend in issue alignment by looking at the correlation between pairs of issues over time. Unfortunately, since 47 issues generate 1,081 potential pairs, we cannot show the trend for all pairs" (p. 430). On the other hand, perhaps this is because political scientists naturally seek to find stable, central attitudes that help citizens understand the more abstract, distant political opinions (Campbell et al., 1960), and this is often the focus of works characterizing public opinion rather than generalized issue constraint across policy domains. But before turning to attitudes that potentially constrain political belief systems, our preliminary concern is whether there has been any change at all in the overall levels of constraint among the mass public, as Converse conceptualized it. We reserve what attitudes might be driving any potential changes in constraint to the second half of this paper.

Much of the preliminary work conducted on this question appears to suggest that Converse's dismal view of the public's understanding of "what goes with what" (p. 9, Converse, 1964; Campbell et al., 1960) was generally correct. Baldassari and Gelman (2008) conduct a large timescale analysis with ANES data with 47 issues to find that from 1972 to 2004, issue attitudes were largely uncorrelated with one another across policy domains. Even when restricting analyses to those issues that were increasingly correlated with partisanship or ideology, the average correlation between any pair of issues was 0.15, and issues that were in the same policy domain only generated a correlation of 0.23 (Baldassarri and Gelman, 2008), a very poor showing for issues that are explicitly related to one another and certainly jives with Converse's original findings (Converse, 1964, pp. 31-32). Kinder and Kalmoe (2017) replicate Converse's findings for constraint among specific beliefs in the 1992 ANES using four domestic issues: defense spending, spending on government services, "aid to Blacks," and provision of government healthcare. The findings are again abysmal. When analyzing average Pearson's r correlations for a broader set of issues from 1972-2012 ANES surveys, the average correlation is a paltry 0.16, with an average correlation increase of 0.01 per decade (Kinder and Kalmoe, 2017). As the authors colorfully point out, if the rate of issue constraint were to continue to increase at this turtle's pace, the mass public will reach elites' levels of constraint by the year 2315.

Is this still the case? More recently, the picture appears to have possibly changed. An extension of the Baldassarri and Gelman (2008) study finds that from 2004 to 2016, the rate of constraint

increased significantly. Simas (2023) extends the Kinder and Kalmoe study and shows their findings may be limited by the time period they analyze, 1972-2012. Rates of constraint among issue positions increased dramatically on the ANES and CES from 2012 to 2020 (Simas, 2023). Hare (2022) finds that idea-elements across a broad range of issues and attitudes are increasingly inter-correlated with each other along what might be called a single ideological dimension around 2012, supporting the theory of partisan conflict extending to many different issue domains (Layman and Carsey, 2002). Whether this can be called something resembling a coherent “ideology” is a matter we attend to later; there is certainly reason to think this is not the case (e.g., Converse, 1964; Kinder and Kalmoe, 2017).⁴

Rather than sophisticated, philosophical ideological thinking, we believe a more plausible explanation for the possible change in constraint is the clarity of elite signals surrounding political conflict and the nature of the issues that characterize political conflict itself. Elites are now more polarized than at any time since the Civil War (e.g., Hare and Poole, 2014). A likely source for such change was the gradual elite polarization and partisan-ideological sorting that occurred as several social movements, the Civil Rights movement chief among them, reshaped American politics (e.g., Carmines and Stimson, 1989; Hetherington and Weiler, 2009). Importantly, though the newer, dominant political cleavage revolves around social issues, that does not necessarily indicate the old political cleavages over economic policies were forgotten; rather, these conflicts were extended and integrated into the new ones (Layman and Carsey, 2002; Hare, 2022). But as the dominant political conflicts have become increasingly organized around hot-button social issues such as race, gender, sexuality, the environment, gun control, and many others (e.g., Hetherington and Weiler, 2009, 2018), determining “what goes with what” has simply become a lot easier for the average citizen (e.g., Carmines and Stimson, 1989).

All Constraint is Not Created Equal

Though there is certainly contestation over the question of overall constraint in mass belief systems, the more recent suggestion from the literature is that constraint has generally increased over time. However, constraint ought not be distributed equally in the population. Assuming that most citizens are constructing most of their opinions as a result of elite cues rather than genuine, in-depth ideological thinking (though there is dispute here), one must ask who is receiving the elite cues most readily and who is able to employ them readily in their belief systems. A long literature suggests that those who are most politically engaged, or aware, are the ones who are best able to connect political beliefs to one another (e.g., Zaller, 1992).

As it pertains to constraint, specifically, it appears that high-information respondents tend to exhibit much more constrained belief systems than their low-information counterparts (e.g., Lupton et al., 2015). High-information citizens are better able to connect their issue opinions with elite cues (e.g., Zaller, 1992). Those who are interested in politics generally seek out more information than those who are uninterested (Prior, 2019), which naturally leads to a higher propensity to recognize the packages of political opinions that ought to go together given elite conflicts. Moreover, high-knowledge respondents tend to exhibit more rigidity in their political opinions, even when

⁴However, see Simas 2023 for discussion.

presented with conflicting information (e.g., Taber and Lodge, 2006).

That there is a difference in constraint among high- and low-knowledge respondents is perhaps not surprising, then. But the open question is what the extent of this difference is. Politics has often been described as abstract and distant to the average citizen (e.g., Converse, 1964). So the theory goes, citizens are much more concerned with their everyday duties and responsibilities, hobbies or leisure, to be concerned with construct coherent yet abstract political belief systems.

Or are they? Converse himself argued that it could plausibly be the case that citizens' belief systems could exhibit just as much constraint as elites via a top-down process of social diffusion. That is, if the masses took to the elite cues being sent. Even if this were the case, such a change would not necessarily represent deep, considered ideological thinking among the public. Rather, the public would simply be mirroring what could ostensibly be considered ideologically constrained elite belief systems (Converse, 1964). Moreover, the changes in the way information is disseminated in mass media, i.e., via television coverage compared to newspaper coverage, helped narrow the gap in political knowledge between less- and more-educated citizens (Jerit et al., 2006), though this effect is in large part reserved for temporally-bounded facts, not general political knowledge (Barabas et al., 2014). With the advent of social media, politics is theoretically at the fingertips of every citizen, but arguably easier to avoid (Prior, 2007). Even so, technological advances have enabled media organizations to profitably target narrower audiences, seen most clearly in the advent of partisan national news outlets on cable and the internet (Prior, 2013). For instance, research on the roll-out of Fox News finds limited effects on political knowledge in 2000 when market penetration was primarily limited to high-education households, but marginal increases in knowledge as Fox News viewership grew more diverse in 2004 (Schroeder and Stone, 2015). Though certainly consumers of partisan news are more interested and engaged in politics broadly, the messages they receive today about "what goes with what" are plausibly more clearly mapped onto partisan and ideological identities due to the increased access to information sources.

In such an information environment, elite cues have the ability to permeate a population much more effectively than in the past. A major source of constraint Converse identified was social constraint via diffusion of "packages" of "what goes with what" from elites to the population; without effective information dissemination, even this basic level of constraint in mass belief systems is unlikely to be realized (Converse, 1964, pp. 8-10). Thus, vast technological changes, including the relative universality of internet access and the iPhone, to name a few, plausibly led to more information dissemination and a recognition of elite cues (Trussler, 2021), even among those who may not be particularly interested in politics (Katz, 1957), which create the necessary conditions for increases in this form of constraint, whether increased constraint is present or not. All that is required for learning "what goes with what" is for "ideological entrepreneurs" to package together groups of issues in a way that is directly accessible, e.g., via social media (Green, 2024).

We demonstrate this change in Figure 1, which represents the weighted proportion of respondents to the ANES surveys from 1960 to 2024 that reported seeing important differences between the two parties as a proxy for elite cue uptake (e.g., Hetherington, 2001). Such a question is quite a low threshold given any perceived important differences between the two parties ought to be counted here. Yet Figure 1 shows the proportion of respondents who saw any significant differences at all

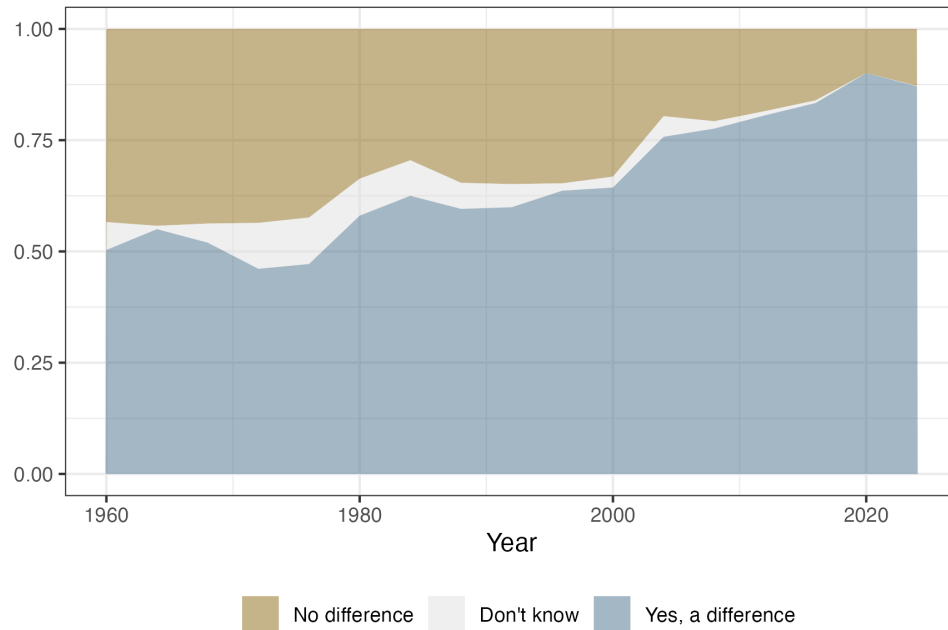


Figure 1: Differences Between Two Major Parties

between the Republicans and Democrats from 1952 to 1976 hovered around a mere 50%.⁵ The Reagan years saw the first uptick in perceived major party differences, and by 2020, this figure topped 90% of the public who perceived major differences, which necessarily includes much of the non-voting public.

Moreover, we subset this trend line by political information using the political knowledge question tapping which party commands a majority in the House of Representatives, the only knowledge question asked consistently from 1960 to 2020. While differences between high- and low-information respondents hovers around 20-30 percentage points from 1960-2000, the information differences attenuate significantly from 2012 onward.

Belief Systems as Belief Networks

Theories of belief systems typically describe their structure as constellations of different attitudes that are more or less connected to one another (Converse, 1964). Empirically modeling this structure is more difficult. Converse opened his 1964 essay with the following qualification: “Belief systems have never surrendered easily to empirical study or quantification. Indeed, they have often served as primary exhibits for the doctrine that what is important to study cannot be measured and that what can be measured is not important to study” (Converse, 1964, p. 1).

However, advances in network analysis allow us to overcome some of these difficulties and empirically represent the structure of belief systems as *networks of beliefs*. Existing studies using

⁵Note: These data collections were conducted directly pre- and post-federal elections, so one would think that perceived differences are inflated during this time period, *ceteris paribus* (e.g., Hillygus and Shields, 2008).



Figure 2: Differences Between Two Major Parties by Information

networks to estimate belief systems have looked at one point in time (e.g., Boutyline and Vaisey, 2017; Brandt and Slegers, 2021). To our knowledge, there is only one political science study that examines temporal change in belief networks (Fishman and Davis, 2022), but it is limited to the 2008 election alone. There has also been some caution given to those estimating network models for smaller samples, among other concerns (de Ron et al., 2022), including modeling belief networks in countries with unstable political systems (Brandt, 2020). As such, we restrict our data to large sample sizes in the United States.

Unfortunately, much of the existing work on issue constraint, sometimes referred to as ideological consistency, is limited by the vast combinations needed to conduct such tests using simple pairwise correlations. Computing alignment between issue pairs quickly balloons results in ways that are not directly representable in article or book formats (e.g., Baldassarri and Gelman, 2008). As such, when correlations are reported, scholars tend to stick to replicating correlation matrices of the sort Converse reported in 1964 (Kinder and Kalmoe, 2017; Simas, 2023). We argue that modeling belief systems as *belief networks* not only allows us to better represent the interdependence among issue attitudes that is more faithful to the underlying Conversion theory of belief systems, but belief network analysis also solves a genuine methodological problem of modeling many different issue-pair correlations and presenting results efficiently.

Data

To calculate changes in constraint in mass belief systems over time, we sought to find a comprehensive survey that includes as many variables as possible over the longest time period possible in order to be able to accurately analyze continuity and change. Thus, we rely on ANES data in con-

ducting the following analyses. The variables from the ANES surveys that we include are restricted to questions that fit two criteria. First, we ask about issue opinions alone. Behavioral questions, such as political participation, are not included. Second, while other variables, such as candidate evaluations, are useful in other contexts, we are interested in how the relative constraint among issue opinions has changed over time, and such opinions are more likely to be ephemeral based on the political context. Because candidate evaluations, for example, change from election year to election year, using this variable would likely introduce error in the estimation of belief systems' change over time. In other words, we place a higher value on the consistency with which opinion questions are asked rather than the completeness of the universe of opinions that could possibly be included in any individual network generated from a single data collection. Third, we sought to use survey questions that are included in as many waves as possible. We begin the data analysis in 1992 because this is the first year that many issue opinions, such as immigration and aid to the poor, are included in the ANES survey.⁶ Beginning our analyses earlier in the time series would significantly decrease the number of issue opinions included consistently on the ANES; beginning our analyses later risks losing coverage of temporal changes.⁷

Thus, our network analyses include an assortment of 15 issue-based opinions: decreasing immigration levels, trade protectionism, the death penalty, affirmative action in hiring and promotion, abortion rights, preventing lesbian/gay discrimination in hiring, increasing defense spending, government-sponsored health insurance, increasing Social Security spending, federal funding for public schools, aid to Blacks, aid to the poor, a federal guarantee for a job and a fair standard of living, and government services and spending.⁸ Again, each of these variables were chosen simply because of their consistent inclusion across the ANES years analyzed in this paper. Other salient political issues, such as gun control, LGBT marriage, or climate change are not included because they were not included consistently from 1992 to 2024.⁹ Many network measures depend explicitly or implicitly on the number of nodes, and we certainly would struggle to meaningfully compare networks made up of dramatically different issues across years.¹⁰

Still, one might reasonably ask whether this particular group of issues could bias our results. There

⁶It is also the first the year that racial resentment and authoritarianism are asked simultaneously, both of which we turn to in later analyses.

⁷Furthermore, the overwhelming suggestion from the literature is that there was not much generalized constraint change before 1992 (e.g., Baldassarri and Gelman, 2008; Kinder and Kalmoe, 2017; Hare, 2022), so we think it reasonable to begin our analyses here.

⁸Including 15 issue opinions is uniquely useful from a network perspective. If we were to compute correlations between each issue pair, this would require $105 \text{ correlations} \times 9 \text{ ANES year observations} = 945 \text{ issue-pair-year correlation combinations}$, which is clearly an infeasible number to display in a coherent and interpretable way.

⁹We focus on presidential-year for the same reason. Presidential-year data collections consistently contain a more fullsome representation of issue opinions. No mid-term-year or other ANES data collection includes all 15 variables, which would be necessary for interpretability.

¹⁰Though work on attitude constraint suggests that looking at additive scales of issue opinions serves to better correct for apparent non-constraint resulting from measurement error (e.g., Ansolabehere et al., 2008), we are limited by what is consistently available on the ANES from 1992 to 2024. However, we are not focused on characterizing constraint at any given point in time as an absolute measure, but rather as a relative measure compared to other years in the ANES data collections. Moreover, because we prioritize consistency in survey questions asked over this time period, our findings for constraint in any given year *relative to other years* can be directly interpretable from a single-item survey questions for a given issue.

are two reasons we believe this to be unlikely. First, the variables included represent a broad range of issue areas. Indeed, the issues included are ones that have been studied in prior research, and the number of issues analyzed over time in this way is generally larger than other studies (e.g., Converse, 1964; Kinder and Kalmoe, 2017; Fishman and Davis, 2022; Simas, 2023), including those using networks to estimate political belief systems (e.g., Fishman and Davis, 2022). Even so, the particular issues included are unlikely to be determinative for our purposes—analyzing the relative constraint within mass belief systems. In Converse’s own words, “The assortment of issues represented is simply a purposive sampling of some of the more salient political controversies at the time of the study... [T]he specific beliefs tested are rather beside the point... [W]e would expect the same results from any other broad sampling of political issues or, for that matter, any sampling of beliefs from other domains... What is generically important... is the difference in levels of constraint among belief-elements” (1964, pp. 31-32).

Network Estimation

Using ANES data from 1992 to 2024, we employ a series of network analyses and evaluate the evidence presented in the context of the existing literature. In it, each issue opinion position collected from the ANES data is included as a vertex, or “node.” The connections between them, or the “edges,” are weighted by the strength of the correlations within the belief network. Each edge, then, can be thought of as the conditional association between any two nodes (opinions).

More formally, we estimate a Gaussian Graphical Model (GGM) to capture the conditional relationships between political opinions as a belief network where nodes represent variables (opinions) and edges represent direct connections between those variables. The edge weights (and therefore structure of the graph) are determined by the inverse of the covariance matrix, or precision matrix, which, after normalization and sign reversal of the non-zero elements, leaves us with a weighted adjacency matrix where edge weights can be interpreted as partial correlation coefficients (Epskamp et al., 2018). Nodes that are not directly connected to one another (i.e., have an edge weight of 0) are assumed to be conditionally independent.

Gaussian graphical models are popular in psychometric analyses and translate well to analysis of belief networks (Boutyline and Vaisey, 2017; Epskamp et al., 2018). In order to estimate the precision matrix, many approaches in psychometrics use the graphical lasso (least absolute shrinkage and selector operator; glasso) algorithm (Borsboom et al., 2021; Briganti et al., 2024; Isvoranu and Epskamp, 2023). Glasso relies on L1 regularization for the estimation of the inverse covariance matrix, allowing the weakest parameter estimates to shrink to zero, while retaining the most influential edges that best characterize the network structure (Epskamp et al., 2018). Regularization is particularly useful in the estimation of the precision matrix. For instance, our network has 21 nodes, which means we would need to estimate 231 parameters for our PMRF. By allowing some of these parameters to shrink to zero if they are less useful in explaining the network structure, the glasso produces a sparser network that is easier to interpret, where each edge can be interpreted as a partial correlation coefficient, and the least important edges are omitted. Many correlations between issue preferences are extremely weak and unstable over time (e.g., Kinder and Kalmoe, 2017). Excluding these edges increases the likelihood that the correlations presented here are true correlations while ensuring greater network stability, if the underlying assumption of sparsity is

met.

The degree of penalization in the regularization process depends on the value of the tuning parameter λ , with higher values of λ resulting in greater penalization; when λ is 0, there is no regularization. There are multiple approaches to hyperparameter selection, most notably EBICglasso, which fits a series of models using progressively larger values of λ and selects the model that minimizes the extended Bayesian information criterion (EBIC). EBIC calculation itself requires the additional specification of a hyperparameter γ that dictates the preference for sparsity. EBICglasso performs well at identifying the strongest edges when the underlying assumption of sparsity is met, at small sample sizes, and particularly with high dimensional data where the number of variables exceeds the number of observations (Williams and Rast, 2020; Isvoranu and Epskamp, 2023). However, when the underlying network is dense, as sample size increases, EBICglasso results in an increasing proportion of false edges surviving regularization (Williams and Rast, 2020; Epskamp et al., 2018).

Instead, we estimate the precision matrix using the model selection algorithm `ggmModSelect` developed by Epskamp et al. (2018). Because our data is ordinal and in many cases skewed, we rely on Spearman correlations to estimate the covariance structure of the variables.¹¹ `ggmModSelect` first uses glasso to estimate 100 precision matrices at 100 non-zero values of λ , before refitting each of the models (retaining only the edges that were not zeroed out in the initial regularized estimation) without regularization and selects the model that minimizes the Bayesian Information Criterion (Epskamp et al., 2018).¹² Then, it re-fits the model iteratively, removing or adding one edge at a time and retaining the changes that result in an improvement in the BIC until no changes would result in improvement. Although `ggmModSelect`, as we implemented it, still relies on glasso for the estimation of the precision matrix, it does not suffer from as dramatic an increase as EBICglasso in falsely included edges as sample size increases, while still maintaining sensitivity (Isvoranu and Epskamp, 2023). The algorithm relies on regularization as a first step in order to speed up estimation, but because the inclusion and exclusion of each edge is then iteratively assessed without regularization, it is typically described as an unregularized estimator (Isvoranu and Epskamp, 2023).

Computing Constraint in Belief Networks

In calculating constraint in belief networks, we seek to be as faithful to the broadly accepted theory of “functional interdependence” as possible (Converse, 1964). To do so, we first use a measure of average path length (APL). APL calculates the average length along the shortest path between all nodes in a network (Warncke, 2024). Prior analysis of belief networks (Warncke, 2024) shows that APL serves to better estimate interdependencies of beliefs compared to average correlational strength (e.g., Converse, 1964; Boutyline and Vaisey, 2017) or leading eigenvalues (Stimson, 1975;

¹¹Though polychoric correlations are perhaps more typical for ordinal data (e.g., Fishman and Davis, 2022), Isvoranu and Epskamp (2023) found using polychoric correlations as input offered little improvement. Rather, doing so induced some decline in the performance of several network estimators using skewed data.

¹²Technically, the algorithm also selects the best model using EBIC, but when $\gamma = 0$, as was our approach and the default, this corresponds to the BIC.

Jessee, 2012).¹³ We follow Opsahl et al. (2010) in calculating the APL for weighted networks where the distance between any two nodes in the network is defined as the sum of the reciprocal edge weights multiplied by a normalizing constant.¹⁴ Shortest paths are computed using Dijkstra's algorithm, and the average distance is taken across all shortest paths in the network. Though our partial correlation networks can (and do) produce negative edge weights, we use the absolute value of the edge weights for the calculation of the average distance because it better captures how closely related two nodes are.

APL is directly interpretable in the sense that issue opinions that are more closely interconnected will, by definition, return a shorter path between any two nodes. Averaging over the shortest path length for all nodes in the network allows us to estimate a global measure of constraint over time, given that the number of nodes and the issue opinions examined in each year are held constant.

However, APL is dependent on the number of non-zero edges present in the network, which is itself a function of sample size. Thus, while APL is relatively more common (e.g., Boutyline and Vaisey, 2017; Warncke, 2024), we also present results for a second measure—network density—that computes constraint as the sum of the absolute edge weights divided by the total *possible* edge weights that could be present in a fully connected network. We report this measure in addition to APL because it does not rely on the existence of edges and has been used in similar research seeking to characterize political belief networks (Fishman and Davis, 2022). While neither measure is perfect, we present both as our current best approximation of a measure of static constraint as Converse defined it.

Results

Figure 3 displays the average path length (APL) for 15 issue opinions consistently asked on the ANES from 1992 to 2024. Figure 4 presents the same results for network density. There are notable results of both constancy and change. There appears to be a moderate decrease in constraint (or increase in average path length) between 1992 and 2008, but there is a large shift from 2008 to 2012, with respondents in 2012 being markedly more constrained compared to 2008. From then on, constraint continues to increase, albeit moderately, through 2024. The same results are mirrored when examining network density (where higher values indicate more constraint and lower values indicate less). The network appears to become quite a bit more dense, or strongly interrelated, between 2008 and 2012 and continues to increase thereafter.¹⁵

To test whether the changes we observe are in any way significant, we conducted nonparametric permutation tests, as proposed and validated by van Borkulo et al. (2023). Each permutation involved pooling the 1992 and 2024 samples, before drawing random samples of respondents without replacement from the pooled data equal to the sample size of the original networks. With these new samples, two new networks are estimated, re-calculating the centrality measures for each node in the network. We then calculate the difference between each centrality measure (the test

¹³See Warncke 2024 pp. 12-13 and 88-92 for simulation tests showing why APL is a superior measure.

¹⁴The mean edge weight in the network.

¹⁵Again, we are still working through how best to characterize these results in light of work on the ANES sample composition changes that occur from 2008 to 2012 and thereafter.

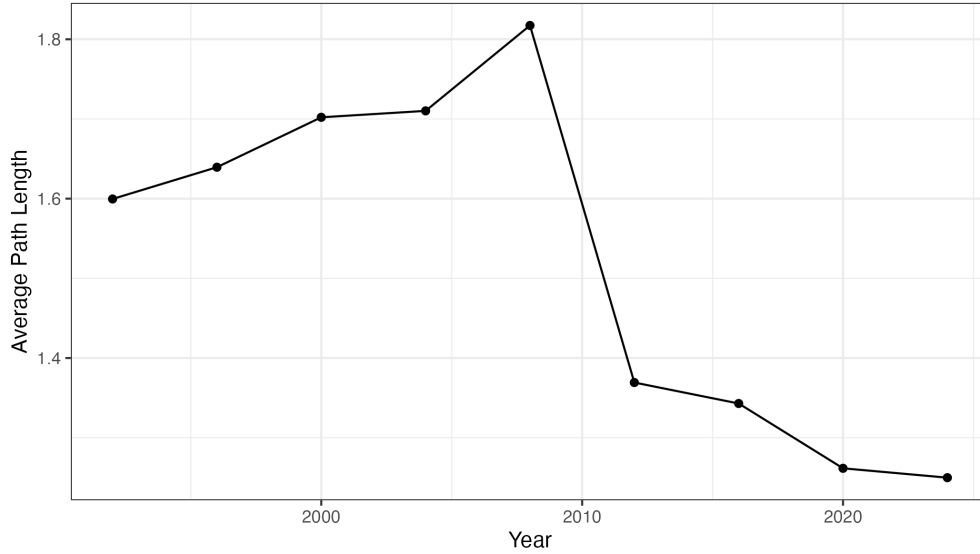


Figure 3: Average Path Length from 1992 to 2024 in Issue Opinion Networks

statistic) and repeat the process 1,000 times to get a distribution under the null hypothesis—that the underlying networks were drawn from the same distribution (that the constraint levels are equal to each other). Table 1 reports the observed difference between each network pair’s APL and network density scores, as well as the p-values for each test.

To assess whether constraint varies by respondent information levels, we re-estimate the networks for each year, separating high- and low-information respondents. We subset each year’s data according to five political knowledge questions that reflect a respondent’s knowledge about temporally-bounded facts.¹⁶ To account for differing levels of difficulty and slightly different question availability across years, respondents were split into high- and low-information groups by percentile. High-information respondents were among the top 30th percentile for correctness across all questions that year, and low-information respondents were among the bottom 30th.¹⁷

Figures 5 and 6 displays these results. In early years, the gap between high- and low-information respondents is substantial: In 1992, low-information respondents are markedly less constrained than their high-information counterparts. These differences significantly attenuate in later years. By 2024, there is no significant difference in constraint between high- and low-information respondents (Tables 2 and 3). While both high-information and low-information respondents appear more tightly constrained from 2012 onward, these differences are not occurring in perfect parallel. Thus, the increased levels of constraint exhibited in Figure 3 can be attributed in large part to low-information respondents rather than high-information respondents.

¹⁶The specific questions used are: correctly identifying the House majority party, correctly identifying the Senate majority party, recognizing the Speaker of the House, recognizing the Chief Justice, and correctly placing Republicans to the right of Democrats on an ideological scale. We use these questions because they are consistently asked across all data collections analyzed from 1992 to 2024.

¹⁷See Figures 16 and 17 in the Appendix for results that split respondents by knowledge levels according to whether they answered *all* knowledge questions correctly (high) or incorrectly (low).

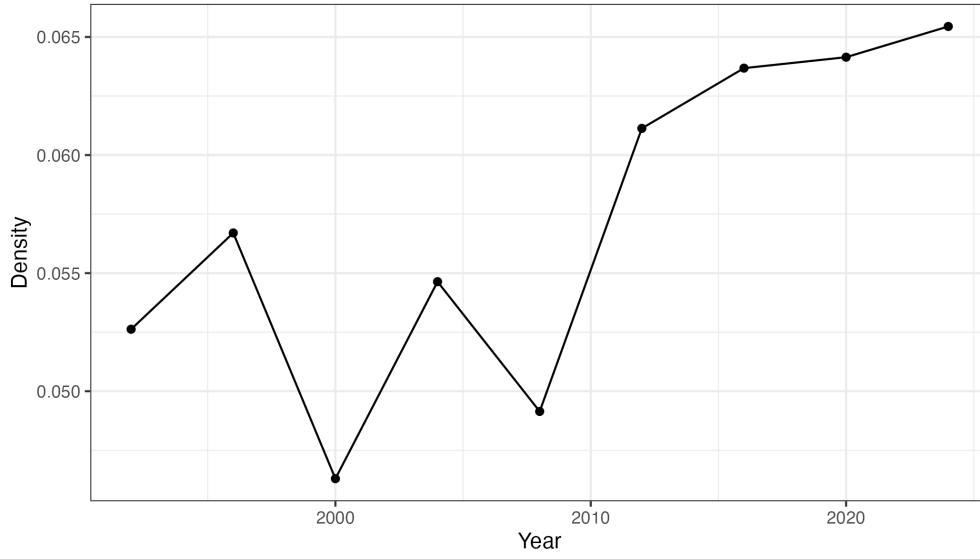


Figure 4: Network Density (weighted) from 1992 to 2024 in Issue Opinion Networks

Table 1: Comparing 1992 and 2024

Measure	1992 – 2024 Difference	p-value
APL	-0.350	0.000
Density	0.013	0.000

Note: The difference is calculated by subtracting the 2024 value from the 1992 value. For APL, negative values denote decreased constraint while positive values indicate increased constraint. For density, negative values denote heightened constraint.

Table 2: Comparing Average Path Length by Information

Year	Low Information	High Information	Difference	p-value
1992	2.154	1.621	-0.533	0.000
2024	1.380	1.406	0.026	0.752

Table 3: Comparing Network Density by Information

Year	Low Information	High Information	Difference	p-value
1992	0.038	0.056	0.018	0.000
2024	0.062	0.06	-0.002	0.468

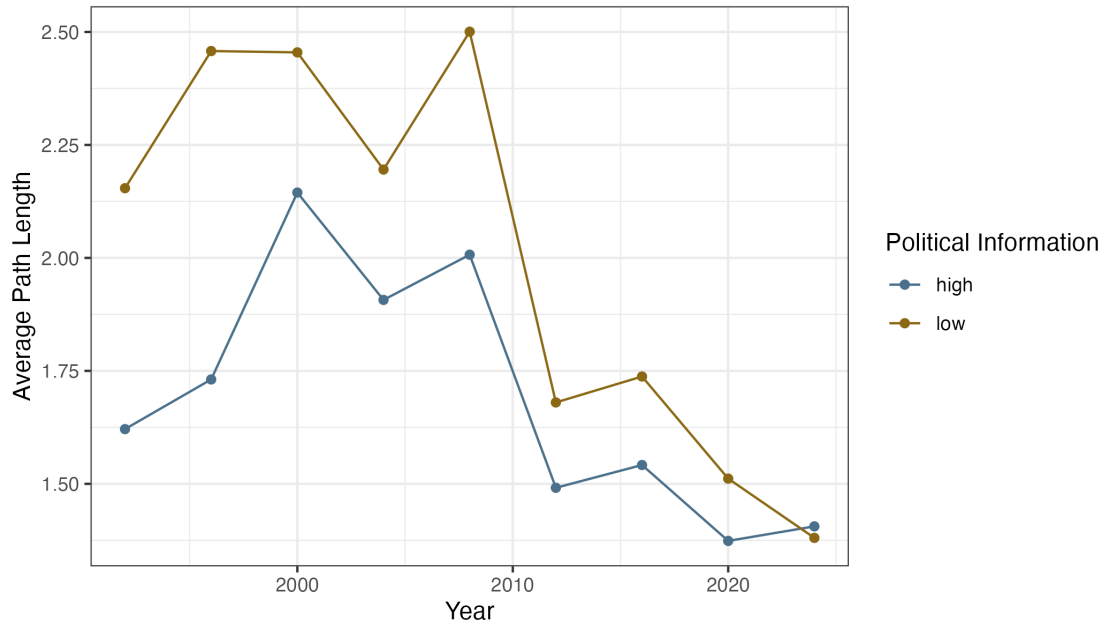


Figure 5: Average path length by Information: 1992 to 2024

Note: Results show network density from 1992-2024 using ANES data collections. Those in the top 30th percentile for political knowledge are considered high-information, while those in the bottom 30th percentile are considered low-information.

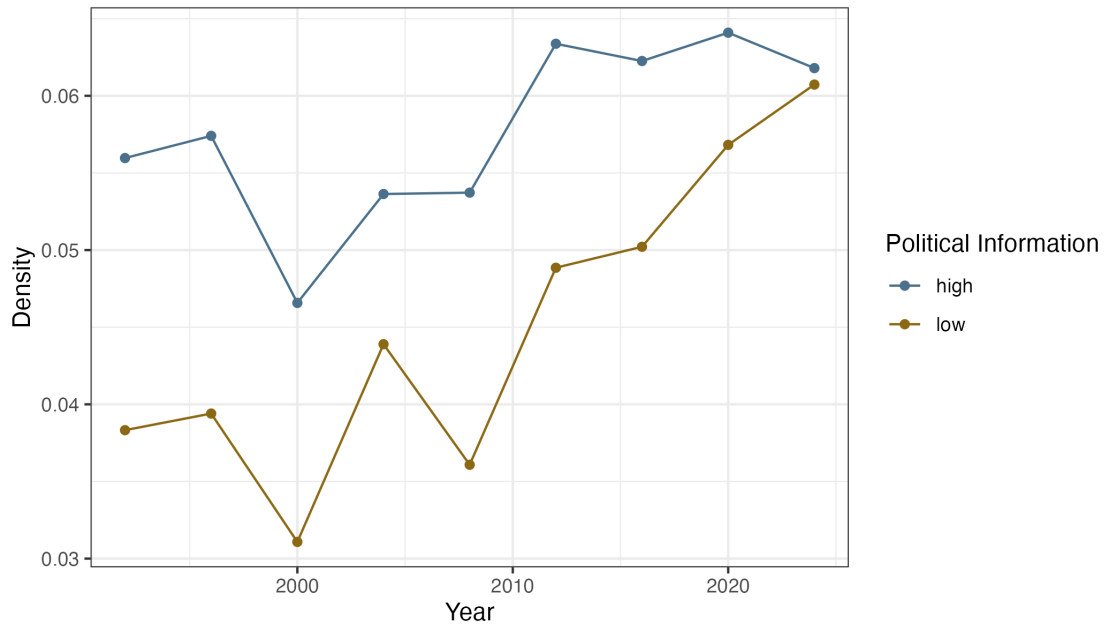


Figure 6: Network Density (weighted) by Information: 1992 to 2024

Note: Results show network density from 1992-2024 using ANES data collections. Those in the top 30th percentile for political knowledge are considered high-information, while those in the bottom 30th percentile are considered low-information.

Constraining Factors

Our results appear to show that overall constraint has indeed increased quite significantly in the population from 2012 onward, with likely heterogeneity by political information; the question remains which political *attitudes* might be exhibiting constraint on these issue *opinions*, particularly among low-information respondents. Thus, we now turn to the underlying structure characterizing mass belief systems.

To compare network structures across years, we necessarily need to hold the nodes observed constant. Thus, unfortunately, we are in some ways limited to the questions consistently asked on the ANES during the time period analyzed. Plausible candidate attitudes asked consistently from 1992 to 2024 are partisanship, ideology, racial resentment, authoritarianism, egalitarianism, and moral traditionalism.¹⁸

Partisanship

Much ink has been spilled regarding partisanship's relationship to political opinions and attitudes. So much so that it is impossible to fully and accurately recount that literature here. Suffice it to say, that, after much debate in the literature for many decades, the idea of partisanship as a “perceptual screen” through which voters interpret political information and form opinions has only become better substantiated over the years (Campbell et al., 1960; Bartels, 2002). Plenty of evidence suggests that the influence of partisanship has greatly increased over time (e.g., Bartels, 2000, 2002; Green et al., 2004). According to the 2020 and 2024 ANES data collections, “strong” partisans have grown to approximately 44%—just under half of the overall population.¹⁹ Partisan “leaners” largely behave similarly to “strong” or “weak” partisans (Keith et al., 1986; Petrocik, 2009). Economic evaluations are skewed by partisanship (Bartels, 2002).²⁰ While partisan identity, or even strength of partisanship, alone is not everything (Highton and Kam, 2011; Chen and Goren, 2016; Klar et al., 2022; Hetherington et al., 2025; Eveland Jr and Gee, 2025), partisanship itself is a powerful force acting on political opinions.

As it pertains to issue constraint, specifically, partisanship is increasingly correlated with many issue opinions (e.g., Hare, 2022). There is much debate, however, over whether this is a result of partisan sorting (e.g., Levendusky, 2009; Fiorina et al., 2008) or partisan updating of political opinions (e.g., Abramowitz and Saunders, 2008). Relatedly, there is contestation over whether partisanship is the central attitude *causing* changes in political opinions or whether issue opinions themselves exhibit an influence on partisanship (e.g., Highton and Kam, 2011). For the purposes

¹⁸Indeed, 1992 is the first year authoritarianism (and many issue opinions) are measured. 1988 is the first year that racial resentment is measured on the ANES, but because we lose authoritarianism and many issue opinions, we do not present those results here.

¹⁹https://electionstudies.org/data-tools/anes-guide/anes-guide.html?chart=strength_of_pid

²⁰Of course, this is largely a function of the party of the president in power and the partisanship of the survey respondent. In fact, there is plenty of evidence that after an incoming president is elected from the opposite party of the sitting president, economic evaluations nearly immediately polarize along party lines, even as the economy remains relatively stable during the transition period (Cummings et al., 2024). What's more, this effect appears to begin occurring even before the new president is inaugurated: <https://today.yougov.com/topics/economy/trackers/state-of-us-economy?crossBreak=democrat>.

of this paper, we simply seek to test whether partisanship is indeed increasingly constraining mass belief systems via network analysis. If partisanship and issue opinions are mostly a product of sorting, we would not expect partisanship to be particularly central to the belief networks. On the other hand, if partisanship is itself bringing more issues in alignment with one another through the operating force of partisanship, it ought to be increasingly constraining over time.

Even so, a cottage industry within the partisanship literature has highlighted the increasing role of negative partisanship—political behavior and opinion formation driven more by out-party hostility than in-party affection—in shaping both political opinions and electoral choice (e.g., Abramowitz and Webster, 2016). Such attitudes have stable determinants (Abramowitz and Webster, 2018; Bankert, 2022) and are measurably distinct from positive partisanship, even as the latter reportedly remains more common (Bankert et al., 2020; Lee et al., 2022). Even so, the increasing aptitude of respondents to express antipathy toward their own party, while largely behaving similarly to fellow partisans, is a notable development—one that could shape how party identification differentially structures belief systems beyond the traditional 7-pt ANES scale.

Ideology

The question of ideology constraining issue opinions suffers primarily from a contestation over what constitutes an ideology and how it is measured (e.g., Luskin, 1987; Treier and Hillygus, 2009). Indeed, the definitions of ideological labels often changes depending on the political context (e.g., Hetherington et al. 2026). Symbolic ideology, meaning one's own self-identification with conservative or liberal ideological labels (Ellis and Stimson, 2012), is what we now attend to rather than ideological consistency or operational ideology. What is important for our purposes is whether issue opinions are substantially correlated with symbolic ideology or not.

Again, there is significant contestation. Plenty of research finds that ideological terms are not well-understood among the public (e.g., Converse, 1964) and historically infrequently employed (Kinder and Kalmoe, 2017), but among those who identify with an ideology, issue constraint can be relatively strong. Since 2012, however, when Kinder's and Kalmoe's data generally end, the correlations between issue opinions and ideological identification is substantially large, particularly among high-information respondents (Simas, 2023; Kozlowski and Murphy, 2021). Of course, high-information respondents themselves are more likely to identify with an ideology (Converse, 1964; Kinder and Kalmoe, 2017; Simas, 2023). Even so, there is preliminary suggestive evidence that there is an increasing correlation among low-information respondents' issue positions on social and moral issues and their ideological identification (Kozlowski and Murphy, 2021). Indeed, ideological self-identification is linked to moral attitudes such as egalitarianism (e.g., Feldman, 1988) and moral traditionalism (Halliez and Thornton, 2021), though such associations do not necessarily themselves lead to generalized issue constraint (Ellis and Stimson, 2012; Claassen et al., 2015; Knuckey, 2005; Goren, 2005).

Racial Attitudes

Racial attitudes have been shown to affect a wide range of policy positions, racial and non-racial alike (Kinder and Sanders, 1996; Kam and Burge, 2019). Indeed, an interesting area for which

Converse found significantly higher constraint was social issues. Specifically, issues that were explicitly non-race-neutral exhibited greater stability and correlations with other, related issues. This suggests that there might be some cognitive force acting on these social, group-based issues that is not present for relatively race-neutral policies and are easier for respondents to grasp (e.g., Carmines and Stimson, 1989). In some cases, racial attitudes are so strong that they can constrain relatively unrelated issue positions (e.g., Tesler, 2012).

The ANES measure of racial attitudes that has been used most consistently over time is racial resentment (Kinder and Sanders, 1996). Though not without critiques in measurement (Enders, 2021) and applications (e.g., Agadjanian et al., 2023), the relationship between racial resentment and a host of public policies is well-documented, including both racial and non-racial issues (Kinder and Sanders, 1996). Indeed, the racial resentment measure is highly stable over time (Kinder and Sanders, 1996) and is a component of each of the attitudes mentioned above. As elites polarized on racial and social issues, racial attitudes have increasingly become linked to partisanship. Still, some work suggests that racial attitudes are not fully exogenous to other political attitudes and identities. For example, panel data analyses suggest that partisanship exerts as much influence on racial resentment as racial resentment does on partisanship (Engelhardt, 2021). Ideological self-identification is significantly influenced by racial attitudes (Enders, 2021). Moreover, there is clear evidence that racial resentment, specifically, is a socialized attitude. For example, counties that had higher populations of enslaved people in the antebellum era tend to report higher rates of racial resentment among whites in modern times (Acharya et al., 2016).

Authoritarianism

A result—or by some accounts, a cause—of political polarization is that many opinions, behaviors, and worldviews that are non-political in nature nonetheless increasingly overlap with partisanship (Hetherington and Weiler, 2018). Indeed, socialized predispositions toward authority affect a broad range of political attitudes, including those about democracy itself. Key (1961) was among the first to conjecture that attitudes toward parental authority might affect political preferences. It took more than 30 years after Key’s observation for survey researchers to put an appropriate measure on the ANES that tapped authority-mindedness. As the dominant political cleavage evolved from economics to social change in the last decades of the 20th century, the ANES 4-item forced-choice measure of authoritarianism in child-rearing came to be strongly associated with many political attitudes, including support for gay rights, civil liberties, and immigration, both legal and illegal (Hetherington and Weiler, 2009; Hetherington and Suhay, 2011; Hetherington and Weiler, 2018). In addition, it is related to many non-political attitudes, including preferences about sports, beer, and music (Hetherington and Weiler, 2018). Authoritarianism is likely a broad enough pre-political disposition to be a good candidate for explaining divides on opinions regarding the gut-level, “easier” social issues that characterize modern political divides (Carmines and Stimson, 1989) as well as a broad range of other issues that tap authority-mindedness. Indeed, authoritarianism, or authority-mindedness, is very much correlated with non-race-neutral policies, such as immigration (Hetherington and Weiler, 2018), in addition to other social and cultural issues, while remaining causally prior to partisanship (Engelhardt et al., 2021).

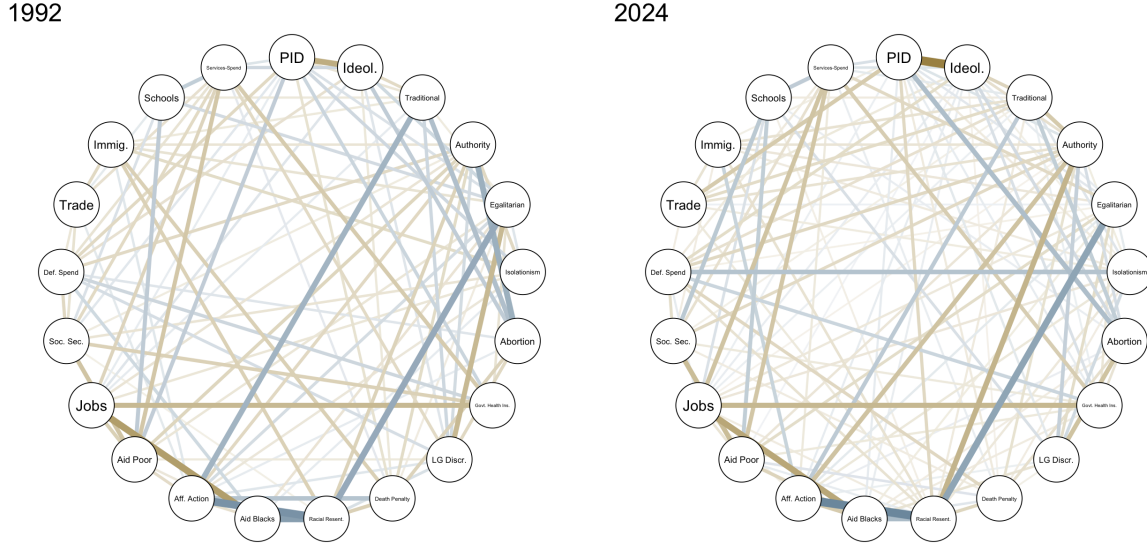


Figure 7: 1992 and 2024 Network Structures

Belief Network Centrality

We now re-estimate the network by including six attitudinal measures—partisanship, ideology, racial resentment, authoritarianism, egalitarianism, and moral traditionalism—in addition to the 15 issue opinions used in the previous network. Doing so allows us to additionally evaluate *which* attitudes are most central to the network and thus exhibit greater influence on other political beliefs. Figure 7 displays the structure of the estimated 1992 and 2024 network models under this specification.

The structure of the networks provides some initial insight into the structure of mass belief systems from 1992 to 2024. Intuitively, party identification, ideology, racial resentment, and authoritarianism each show strong connections with other nodes. Still, the relative explanatory power of each of these variables is difficult to ascertain from the visualizations alone. To investigate how influential each of these variables is, relative to others in the belief networks, we focus on node centrality.²¹

Much as we did for average path length and density, we calculate centrality using the absolute edge weights. Though the signs of the edge weights provide important information about the relationship between any two nodes, the absolute edge weights provide a clearer picture of the strength of those relationships. Most centrality measures developed or adapted for weighted networks require positive weights, and those that do exist with signed analogues often conceptualize a negative weight as indicating a degree of aversion (Opsahl et al., 2010). Our aim in focusing attention on node centrality is to characterize the relative influence of each node. As such, we rely solely on the absolute weights of the edges in the network to compare the strength of positive and negative edge weights when estimating node centrality.

²¹To reiterate, this is a measure of *network centrality*, not centrality in the Converseian sense. The networks presented here can be best described as approximating Converse’s conceptualization of “static constraint” (Converse, 1964).

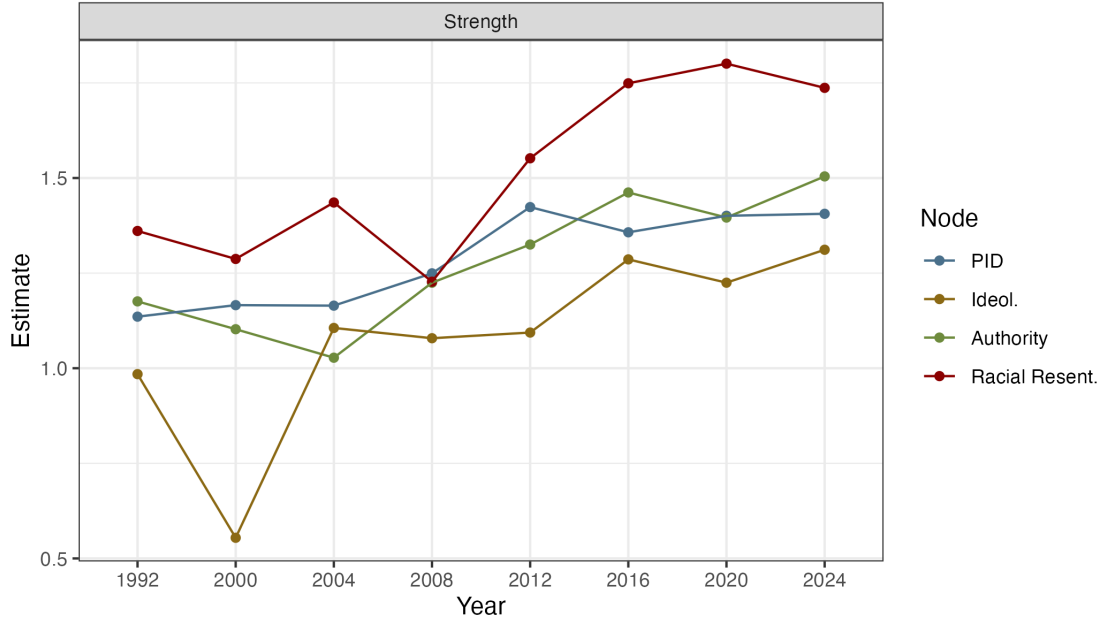


Figure 8: Strength Centrality from 1992 to 2024

Strength centrality—the measure we focus on—is the sum of the absolute edge weights connected to a node and is the weighted equivalent of degree. It is straightforward in interpretation and the most consistent of the network centrality estimates (Isvoranu and Epskamp, 2023). We can think about strength centrality as indicating how closely connected a variable is to the rest of the nodes in the network. Because strength centrality is calculated not only based on how many direct connections an attitude has to others, but also the strength of those connections, one or two incredibly strong relationships could potentially outweigh many weak ones. Additionally, strength centrality depends exclusively on the estimated edge weights, which makes it more insulated from the influence of weak edges—a weak connection or small weight cannot disproportionately elevate a node’s strength.

Results

Figure 8 plots strength centrality for the main four attitudinal nodes in our network: partisanship, ideology, racial resentment, and authoritarianism.²² We see some interesting notes of constancy and change. Ideology, party identification, and authoritarianism are fairly similar to one another in any given year (with the exception of 2000, when ideology becomes isolated in the network). Though each attitude exhibits a slight upward trend in strength centrality, the greatest movement comes from racial resentment, whose centrality peaks in 2020 before declining slightly in 2024.

We compare centrality in 1992 and 2024 directly in Figure 9 for all nodes in the network. Racial resentment, authoritarianism, partisanship, and ideology all exhibit statistically significant increases in centrality between 1992 and 2024. The political values measures, egalitarianism and moral traditionalism, are both relatively central to the networks, but to a much lesser extent. Moreover,

²²See Figure 18 in the appendix for full results including egalitarianism and moral traditionalism.

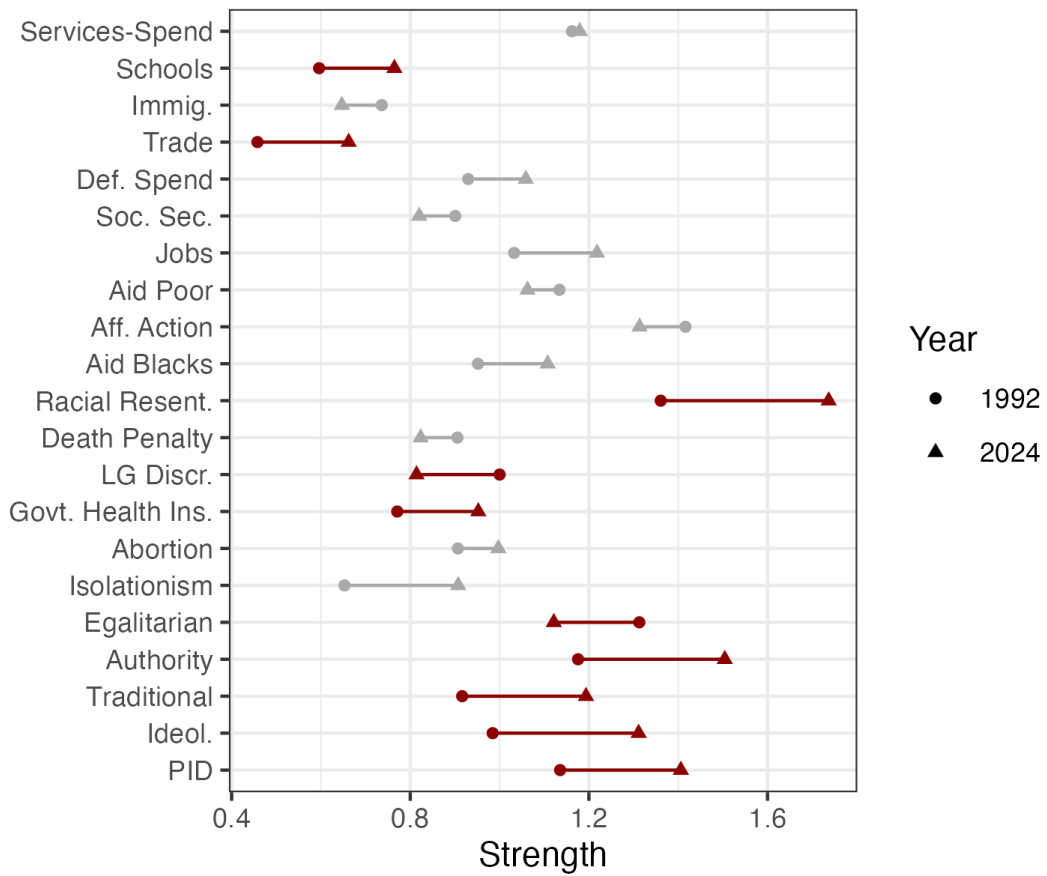


Figure 9: Comparing Strength Centrality Between 1992 and 2024

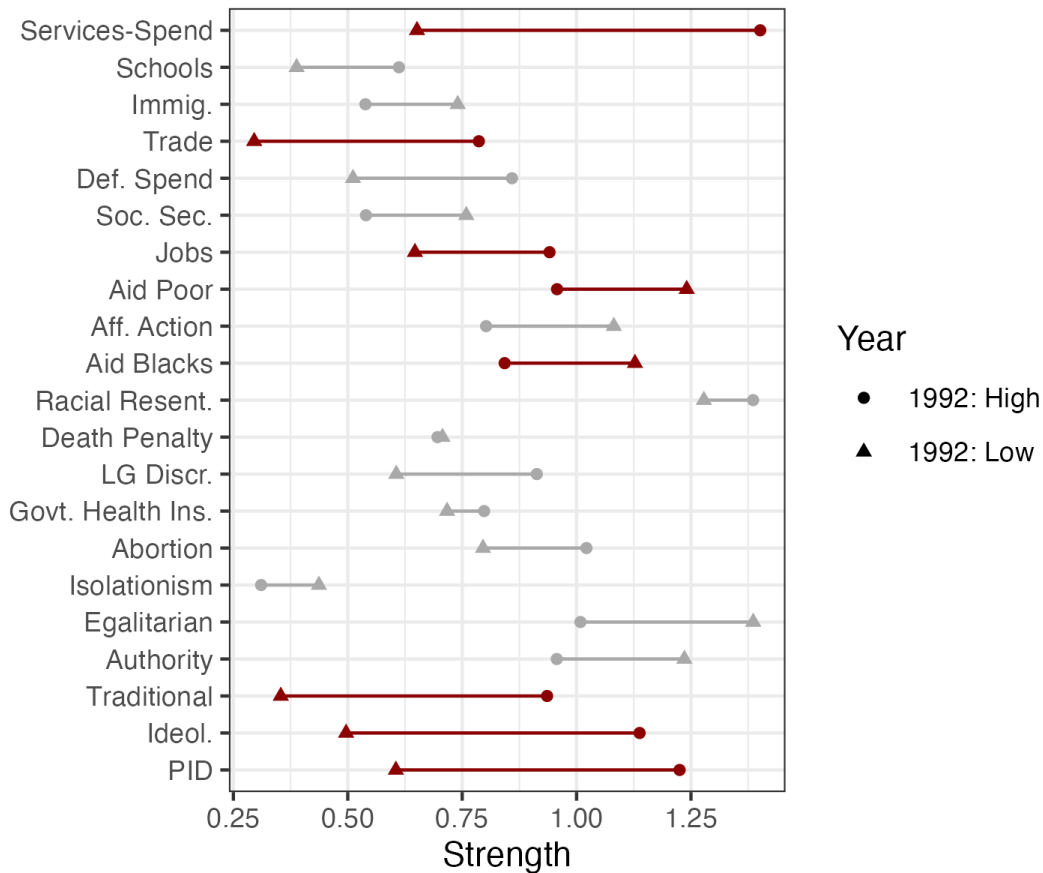


Figure 10: 1992 Strength Centrality Differences by Information

while moral traditionalism increases in centrality from 1992 to 2024, egalitarianism decreases.²³

We investigate these results further by again subsetting by political information. For these analyses, we focus on the beginning and end of our time series: 1992 and 2024. To compare strength centrality between high- and low-information respondents, we again conduct permutation tests, comparing high- and low- information respondents within the same year. Figure 10 plots the strength centrality of each node for high- and low-information respondents in 1992, with significant differences highlighted. In 1992, ideology, partisanship, services-spending, and moral traditionalism are markedly more central among high-information respondents compared to low-information respondents. Indeed, these attitudes (with the exception of moral traditionalism) are among the most central nodes in the high-information network, alongside opinions on services and spending and racial resentment, the latter of which exhibits strong centrality across knowledge groups.

Figure 11 plots strength centrality for 2024 between high- and low-information. The differences between high- and low-information networks are far more muted in 2024 compared to 1992. Of the key attitudinal variables, the only node that exhibits a significant difference is ideology, which is

²³We attribute affirmative action's relatively high centrality to its close association with the racial resentment measure.

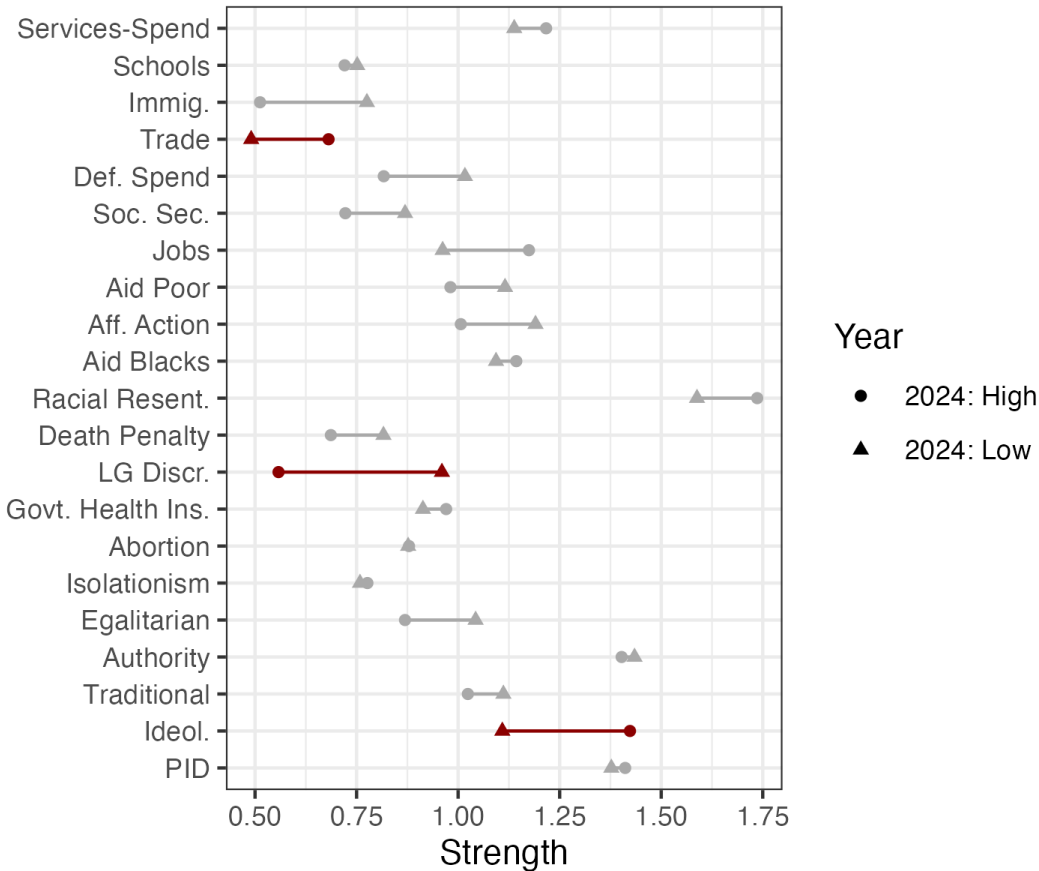


Figure 11: 2024 Strength Centrality Differences by Information

more central in the high-information networks compared to the low-information networks. Racial resentment, authoritarianism, and partisanship remain highly central in both networks, with no significant difference across knowledge groups.

Discussion

Our findings contribute to the literature on the organization of mass belief systems in several meaningful ways. By estimating networks of beliefs, we gain useful leverage in modeling the theorized “functional interdependence” that characterizes belief systems (Converse, 1964). Such a method allows us to overcome methodological problems of estimating large numbers of pairwise correlations between issue pairs and calculating average correlations noted by others (e.g., Baldassarri and Gelman, 2008; Kinder and Kalmoe, 2017). Thus, network analyses are useful in estimating belief system structures. This paper is unique in this literature in using network analysis to 1) calculate constraint among many issue opinions simultaneously and 2) represent results in an easily interpretable manner.

Our results suggest an interesting bridge between conflicting factions in the field regarding issue constraint in mass publics. On one hand, analyzing issue constraint from 1972 to the early 2000s

suggests a low level of constraint that is constant over time (Kinder and Kalmoe, 2017; Baldassarri and Gelman, 2008). On the other hand, recent work conducted on this question suggests that overall constraint among issue opinions has been increasing recently (Simas, 2023; Hare, 2022; Kozlowski and Murphy, 2021). Our results suggest that efforts in these studies to generalize findings about the American public are limited by temporal changes in the mass public's constraint that occurred between the 2008 and 2012 elections. Since 2012, constraint has continued to increase, though only at a marginal rate. However, at this time, we are unable to distinguish this effect from issues of survey composition that could be biasing our results (Hillygus et al., 2017).

Finally, our findings suggest that racial resentment, authoritarianism, and partisanship are the strongest constraining forces on the structure of mass belief systems, and their influence has only increased over time. Symbolic ideology has also increased in its strength centrality from 1992 to 2024, though it is slightly less central than these other measures. While we do not necessarily attribute this to increased ideological thinking, but rather an increased understanding of what labels go with what opinions, future studies can further probe this question. What appears to be the case is that the mass public is not quite as ideologically innocent as recent research has suggested. Indeed, comparisons in strength centrality show that the differences in high- and low-information respondents' readiness to use political concepts such as ideology and partisanship significantly attenuates over time, though more so for the latter than the former. Racial resentment and authoritarianism, on the other hand, exhibit similarly high levels of strength centrality among low- and high-information respondents alike in 2024.

This study also has important limitations. We are primarily limited by the ANES data itself. We sought to characterize constraint over as broad a time period as possible while keeping the specific opinions and attitudes we measured constant over time in order to directly interpret constraint over time. That limits us to beginning in 1992, when several issue opinions and the child-rearing authoritarianism began being asked. Future studies can extend these analyses by looking at alternative opinions or time periods. Such analyses might yield marginally different results, though the coverage and scope of issue opinions covered here lead us to believe this is unlikely to be the case.

A related limitation is that our analyses rely on ANES presidential-year data collections due to the consistency of opinions and attitudes asked on those surveys. However, election cycles can influence how the mass public thinks about what political beliefs go with what. As such, the absolute levels of constraint observed here may be inflated, though relative constraint may still be characterized. Additionally, our measures of constraint are the best that we are currently aware of, but we may be able to find better measures in the future that more closely model the "functional interdependence" of opinions within belief networks.

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Appendix

Average distance and density including attitudinal variables

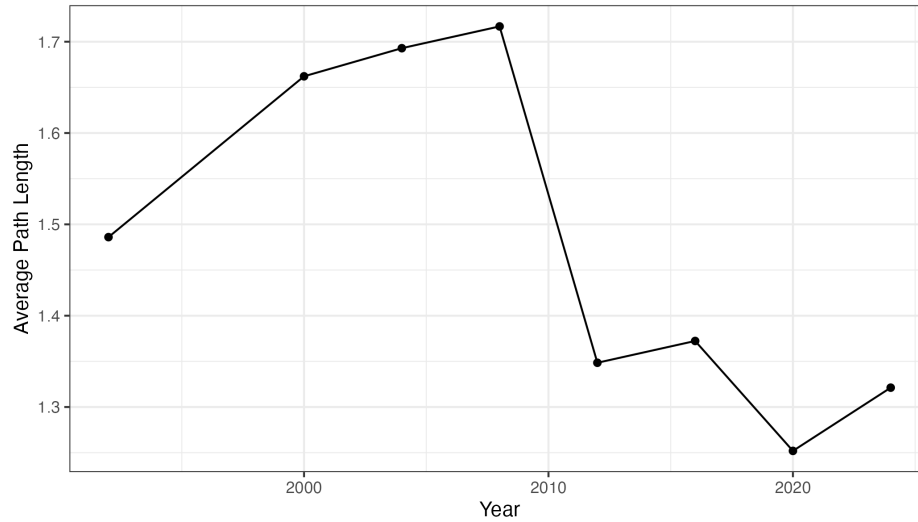


Figure 12: Average Path Length in Full Networks

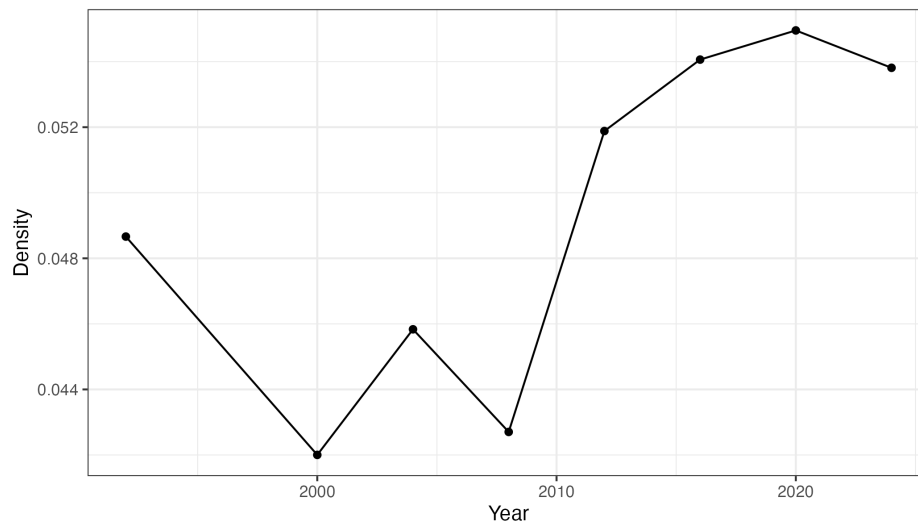


Figure 13: Density in Full Networks

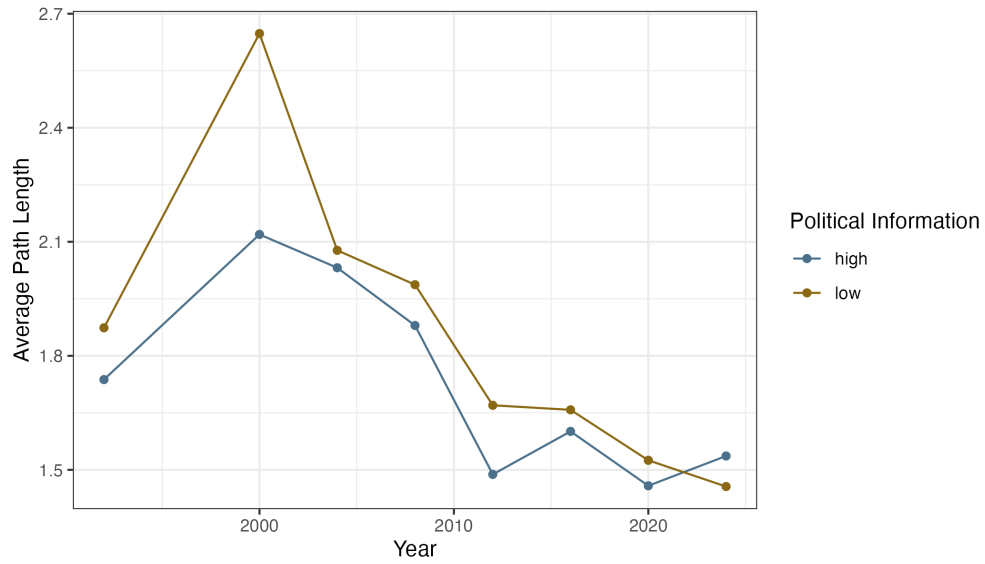


Figure 14: Average Path Length in Full Networks by Knowledge

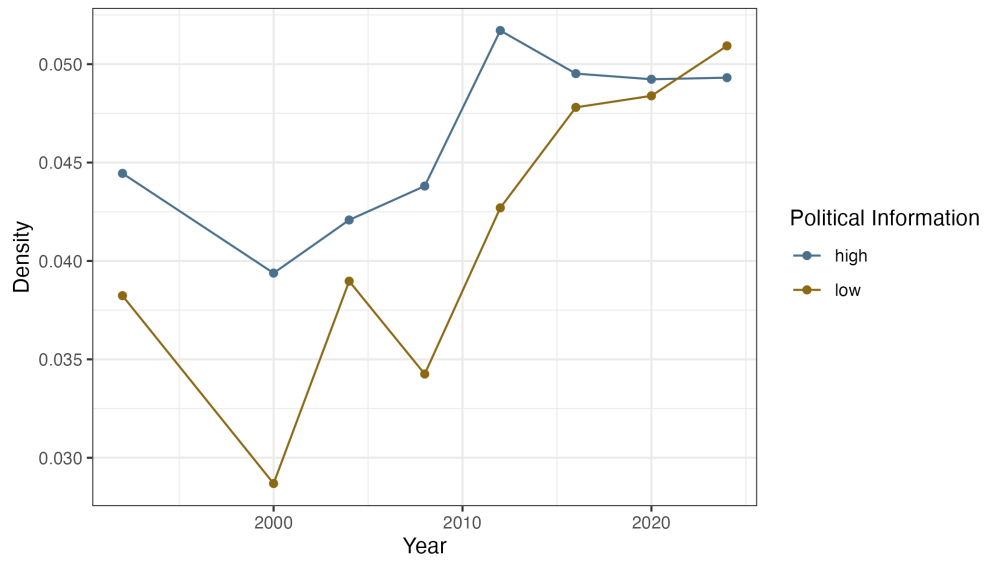


Figure 15: Density in Full Networks by Knowledge

Average distance and density at different knowledge cutoffs

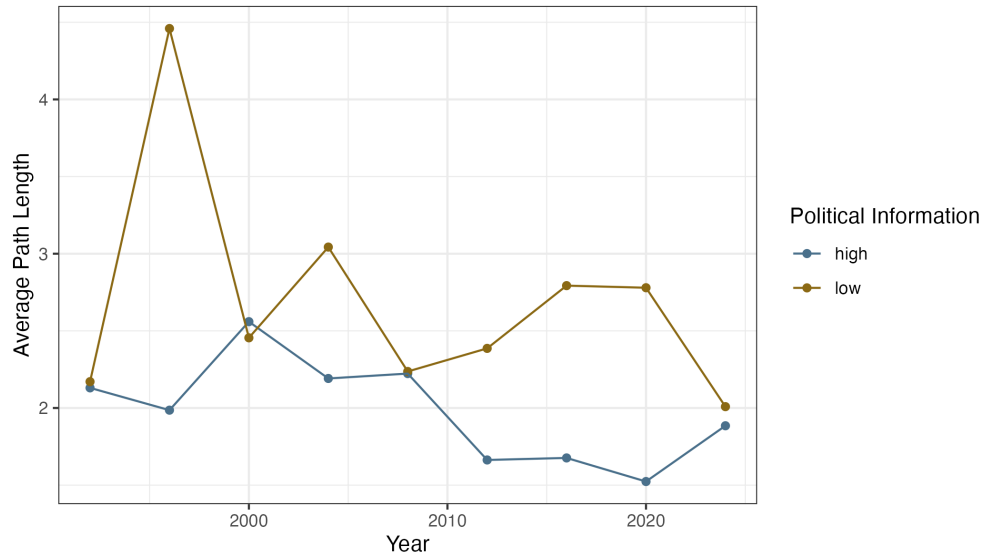


Figure 16: Average Path Length in Issue Networks by Knowledge

Note: Results show network density from 1992-2024 using ANES data collections. Those who answered all 5 knowledge questions correctly are considered high-information, while those who answered none correctly are considered low-information.

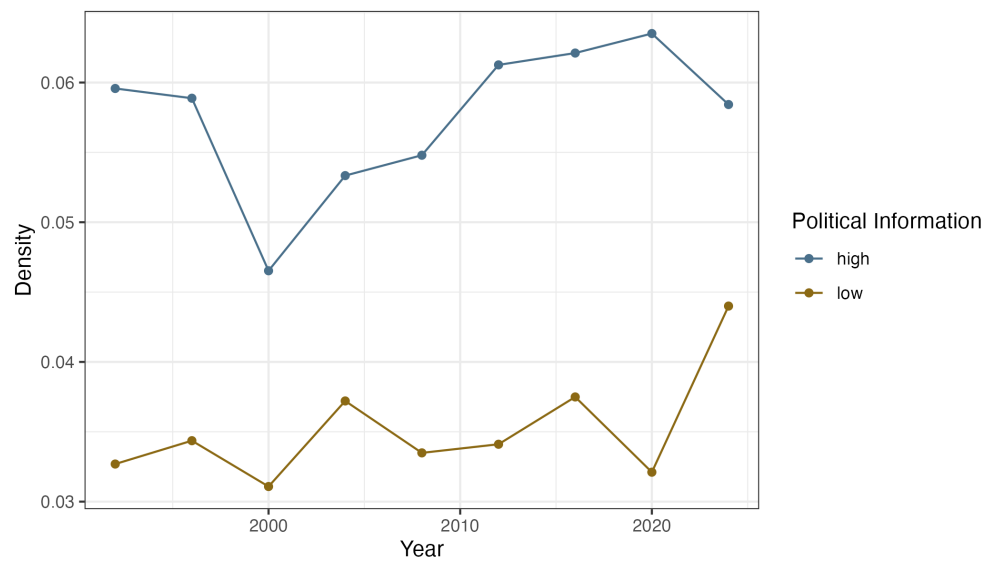


Figure 17: Density in Issue Networks by Knowledge

Note: Results show network density from 1992-2024 using ANES data collections. Those who answered all 5 knowledge questions correctly are considered high-information, while those who answered none correctly are considered low-information.

Strength Centrality

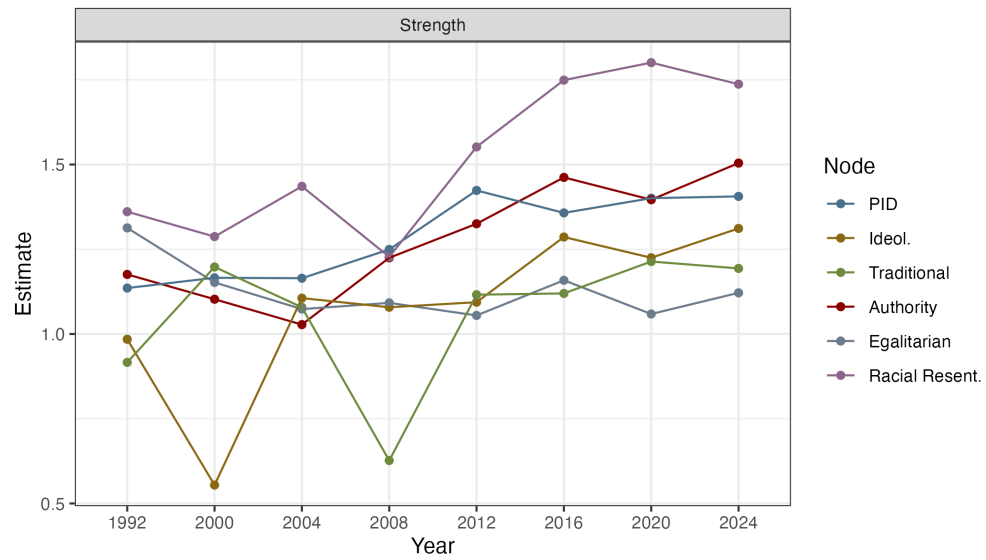


Figure 18: Strength Centrality from 1992-2024

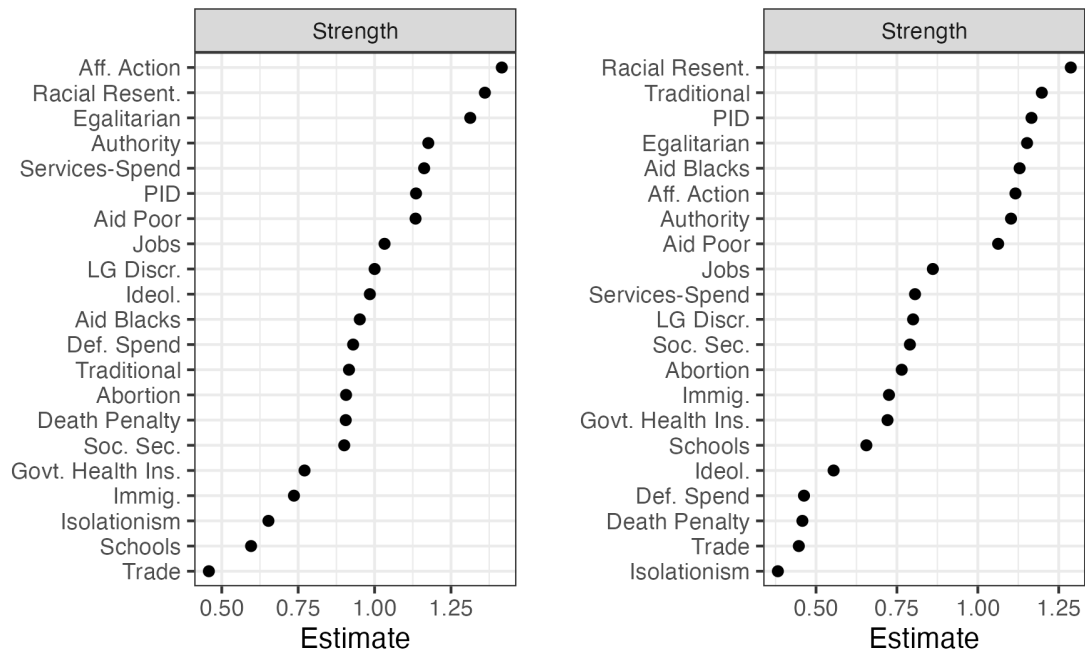


Figure 19: Centrality for 1992 (left) and 2000 (right)

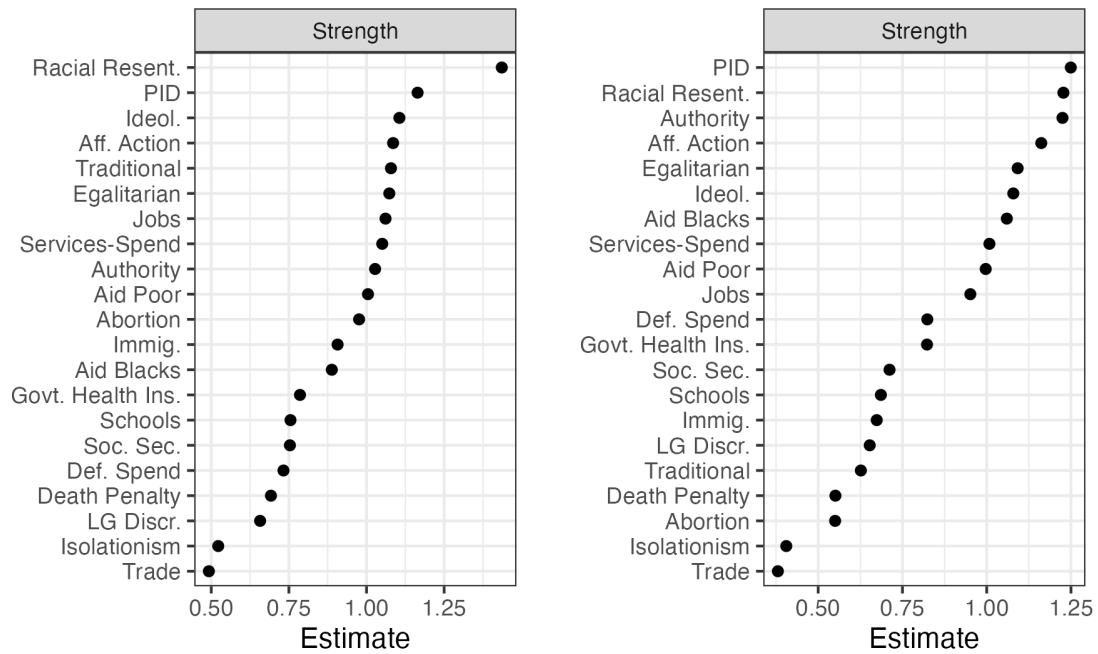


Figure 20: Centrality for 2004 (left) and 2008 (right)

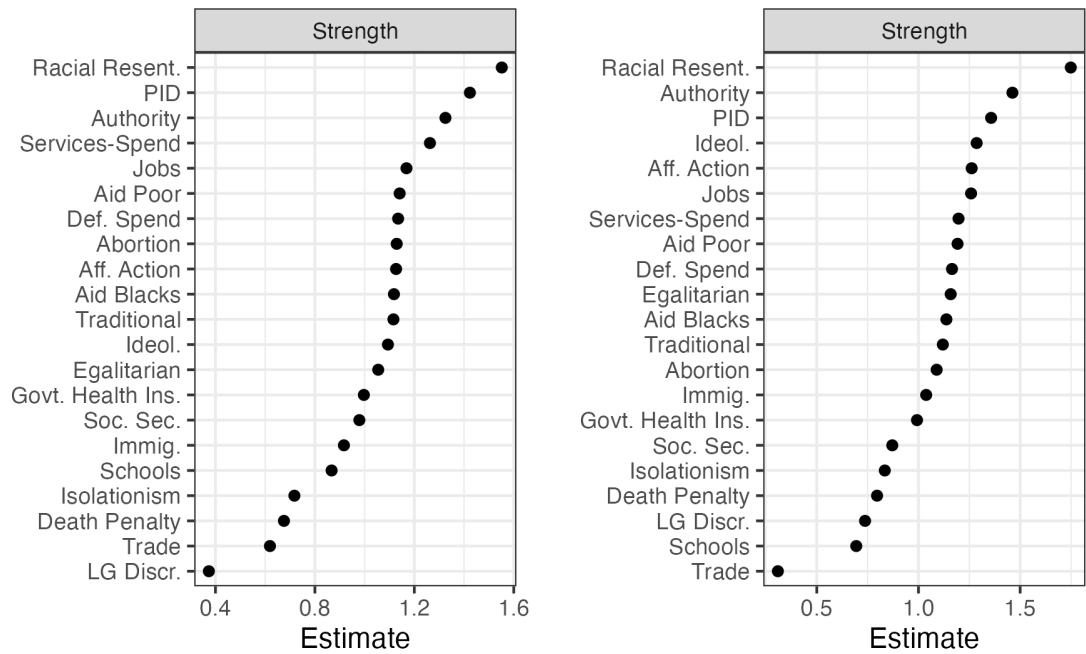


Figure 21: Centrality for 2012 (left) and 2016 (right)

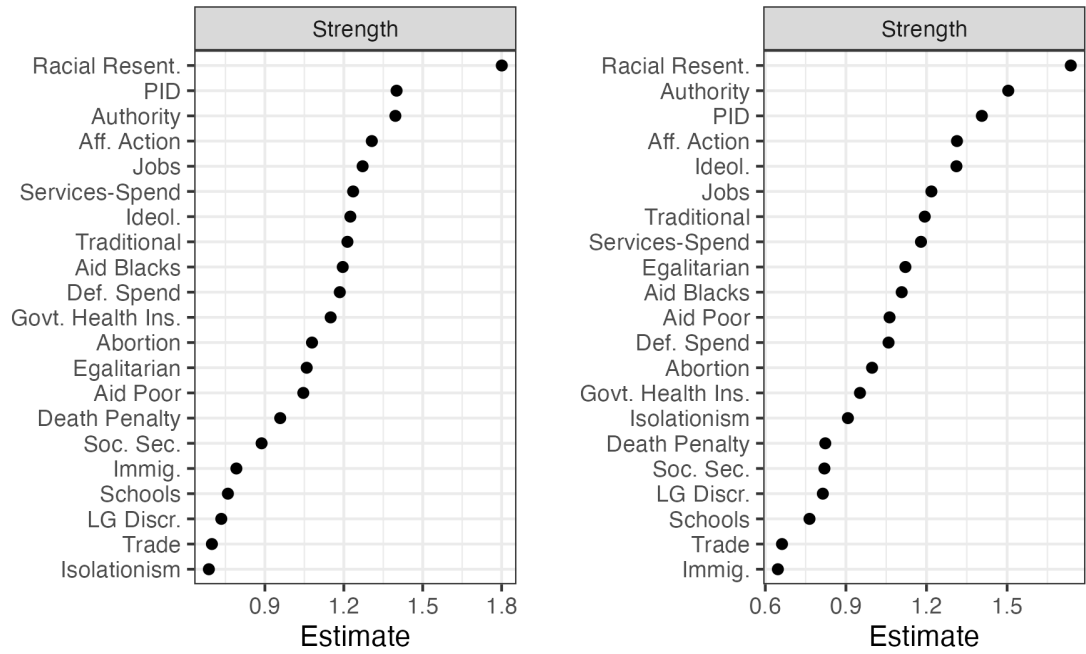


Figure 22: Centrality for 2020 (left) and 2024 (right)